

GenAI Cookbook

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- GenAI Cookbook
- Retrieval-augmented generation (RAG)

TLDR; this cookbook and its sample code will take you from initial POC to high-quality production-ready application using [Mosaic AI Agent Evaluation](#) and [Mosaic AI Agent Framework](#) on the Databricks platform.

The Databricks Generative AI Cookbook is a definitive how-to guide for building *high-quality* generative AI applications. *High-quality* applications are applications that:

1. **Accurate:** provide correct responses
2. **Safe:** do not deliver harmful or insecure responses
3. **Governed:** respect data permissions & access controls and track lineage

Developed in partnership with Mosaic AI's research team, this cookbook lays out Databricks best-practice development workflow for building high-quality RAG apps: *evaluation driven development*. It outlines the most relevant knobs & approaches that can increase RAG application quality and provides a comprehensive repository of sample code implementing those techniques.

! Important

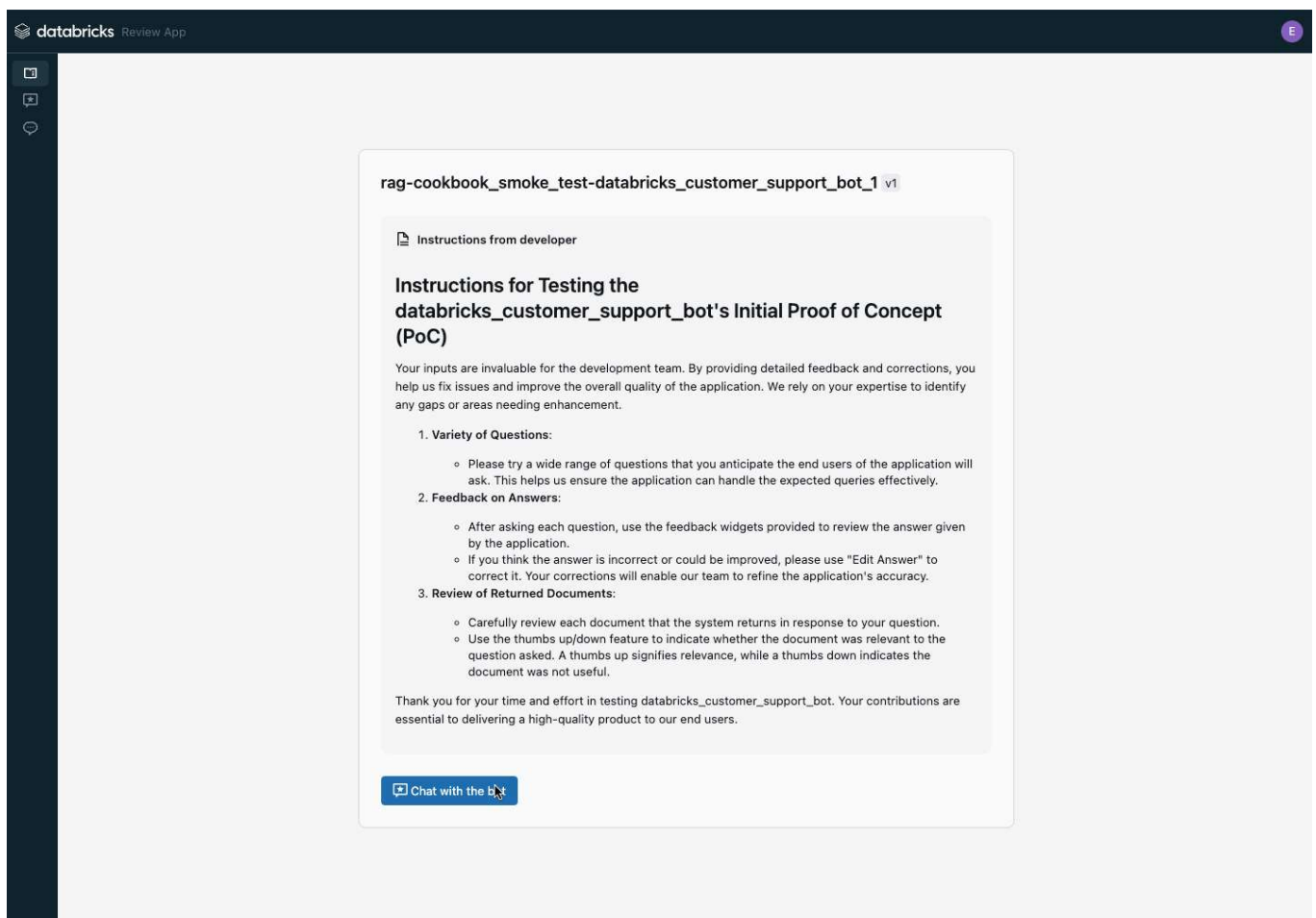
- Only have a few minutes and want to see a demo of Mosaic AI Agent Framework & Agent Evaluation? Start [here](#).
- Want to hop into code and deploy a RAG POC using your data? Start [here](#).
- Don't have any data, but want to deploy a sample RAG application? Start [here](#).

Databricks approach to quality

Fast, code-first developer loop to rapidly iterate toward quality.

Easy to collect human feedback.

Rapid and reliable measurement of app quality.



This cookbook is intended for use with the Databricks platform. Specifically:

- [Mosaic AI Agent Framework](#) which provides a fast developer workflow with enterprise-ready LLMops & governance
- [Mosaic AI Agent Evaluation](#) which provides reliable, quality measurement using proprietary AI-assisted LLM judges to measure quality metrics that are powered by human feedback collected through an intuitive web-based chat UI

This first release focuses on retrieval-augmented generation (RAG). Future releases will include the other popular generative AI techniques: agents & function calling, prompt engineering, fine tuning, and pre-training.

The RAG cookbook is divided into 2 sections:

1. **Learn:** Understand the required components of a production-ready, high-quality RAG application
2. **Implement:** Use our sample code to follow an evaluation-driven workflow for delivering a high-quality RAG application

Code-based quick starts

Time required	Outcome	Link
 30 minutes	Sample RAG app deployed to web-based chat app that collects feedback	RAG Demo
   2 hours	POC RAG app with <i>your data</i> deployed to a chat UI that can collect feedback from your business stakeholders	Build & deploy a POC
  1 hour	Comprehensive quality/cost/latency evaluation of your POC app	- Evaluate your POC - Identify the root causes of quality issues

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